COMP 4446 / 5046
Lecture 4: Inference – Greedy and Search

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THE UNIVERSITY OF SYDNEY

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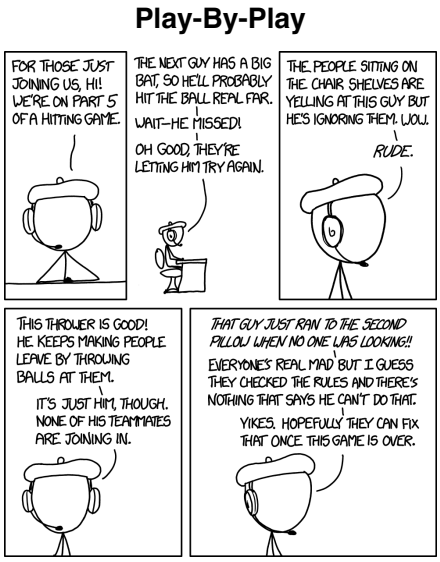

[The thrower started hitting the bats too much, so the king of the game told him to leave and brought out another thrower from thrower jail.]

Source: https://xkcd.com/1593/

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Now, let's explore different inference methods

Data	Examples of the language phenomena we want our system to handle
Model	A function that maps (input, output) pairs to scores
Inference Method	A way to make a prediction for an example given a Model
Metric	A function that gives a score to the output produced by a Model given some Data
Learning Method	A way to update a Model given Data a Metric and an Inference Method

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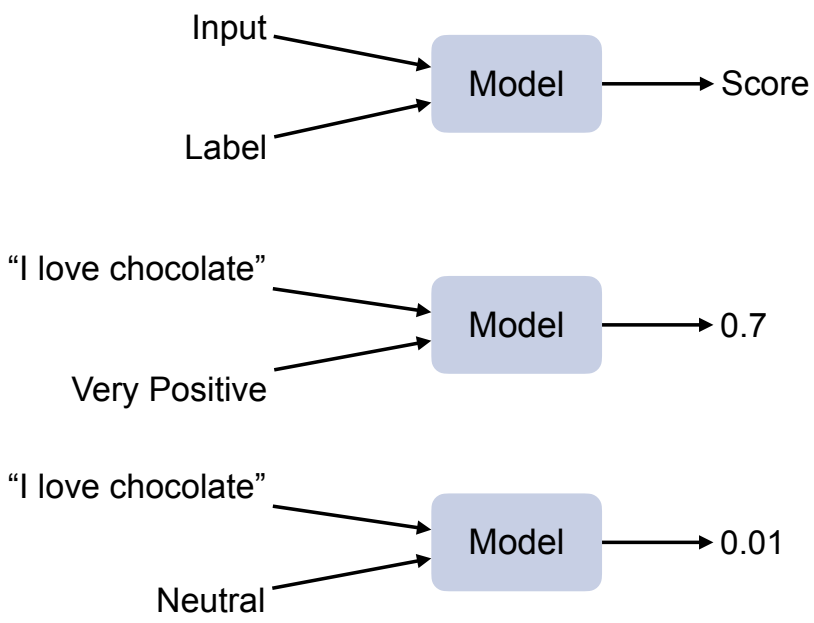
We'll treat the model as a black box



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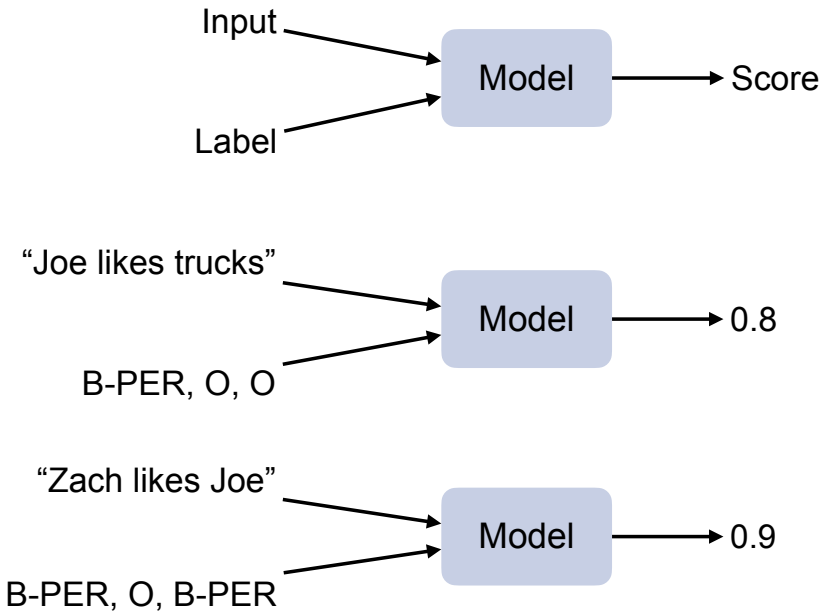
The model could score whole sequences



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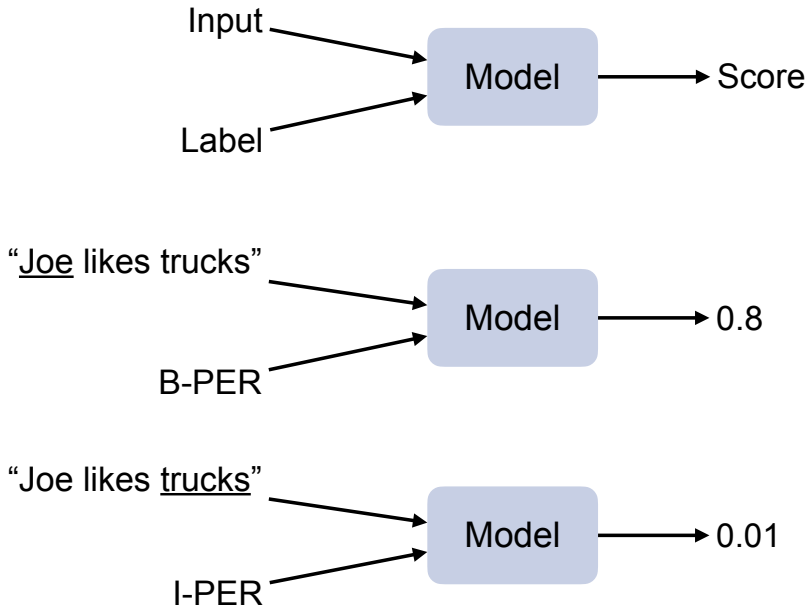
The model could score whole sequences, or just one part



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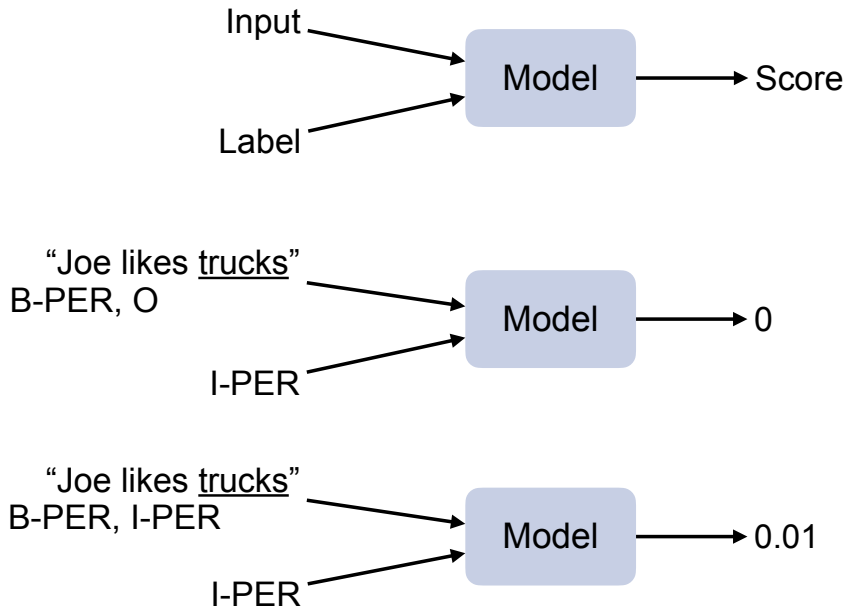
The model could score whole sequences, or just one part, possibly with context



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We'll use several running examples in this section

Noun	Verb	Prep	Noun
Adj	Noun	Verb	Noun
Fruit	flies	like	bananas

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We'll use several running examples in this section

Ingredients:

4oz Chocolate, 70% cocoa
1cup Milk
Chocolate shavings

Steps:

1. Melt chocolate on stove
2. Slowly add milk
3. Heat until simmering
4. Take off heat and let cool completely (~20 min)
5. Return to stove and heat to desired temperature
6. Top with chocolate shavings

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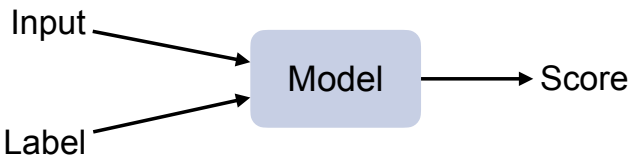
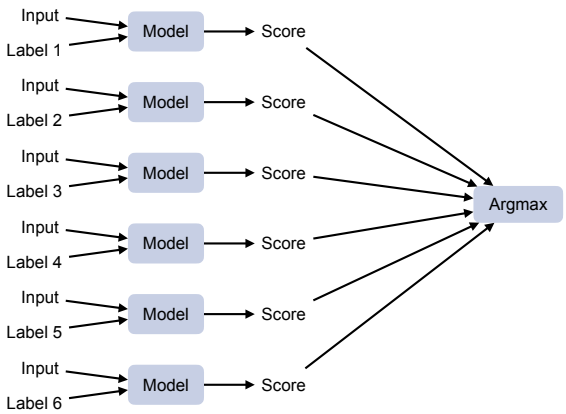
In assignment 2, we are using an exhaustive method





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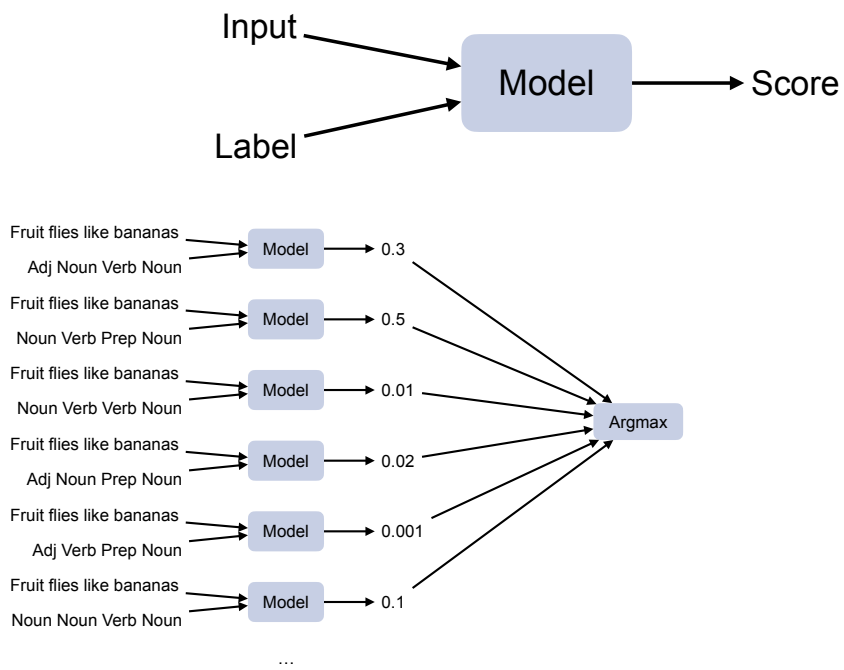

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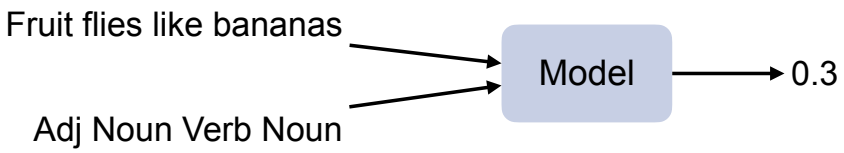
Exhaustive search is flexible, but not scalable

Benefit - The model can look at the entire structure



Problem - for many tasks, the search space is huge

17 tags in Universal Dependencies

Options = $|\text{tags}|^{|\text{words}|}$

For this example, 17^4 = 83,521 options!

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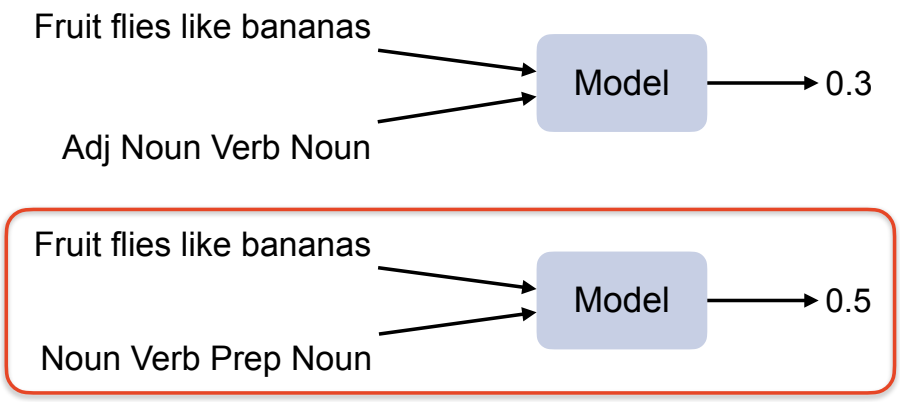
Note: we are looking for the highest scoring output, which might be wrong



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This suggests another way of describing these two components of an NLP system

Model — Tries to assign the highest score to the true output

Inference Method — Tries to find the output that gets the highest score from the Model

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Greedy

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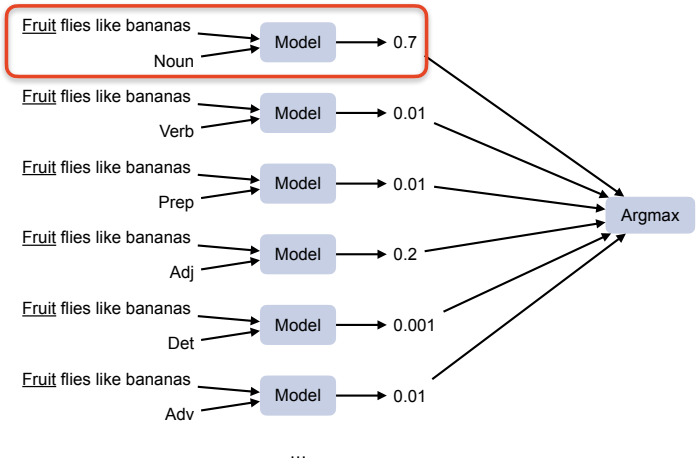
Core greedy idea: Make choices one at a time



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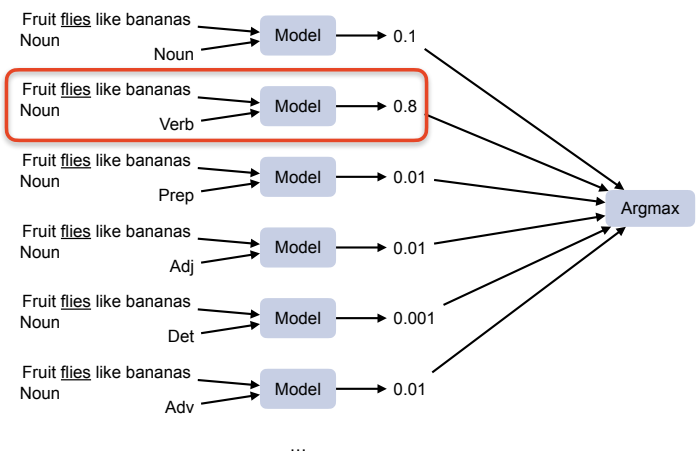

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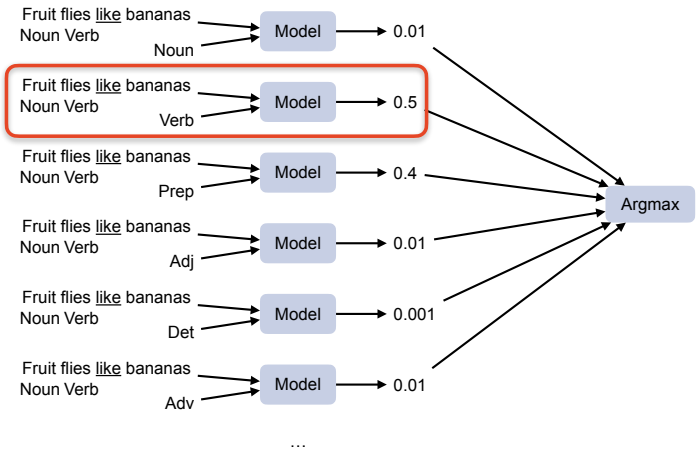
Core greedy idea: Make choices one at a time



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The complexity problem is fixed!

17 tags in Universal Dependencies

Options considered in each step = |tags|

Step = |words|

Complexity = |tags| * |words|
= 17 * 4
= 68

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But… the answer was different

Exhaustive:	Noun	Verb	Prep	Noun
Greedy:	Noun	Verb	Verb	Noun
	Fruit	flies	like	bananas

Note - both of these are wrong, but one is the highest scoring according to the model and the other is not.

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A few notes on comparing exhaustive and greedy

Sometimes the answer can match. For example, if every part of the output is independent.

Greedy has less information about the output, but can still use a lot of context to make the decision

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First variant: **Top-1**

Method: At each step, choose the highest scoring option

This is what we just saw!

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First variant: **Top-1**

Method: At each step, choose the highest scoring option

Ingredients:

4oz Chocolate, 70% cocoa
1cup Milk

Steps:

1. Melt chocolate on stove
2. Slowly add milk
3. Heat until simmering
4. Melt chocolate on stove
5. Add milk
6. Heat
7. Melt chocolate on stove

…

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Second variant: **Random sampling**

Method: At each step, choose using random sampling from the probability distribution

Ingredients:

4oz Chocolate, 70% cocoa
1cup Milk

Steps:

1. Melt chocolate on stove
2. Slowly add milk
3. Heat until simmering

Melt chocolate on stove

Take off heat and let cool completely (~20 min)

0.3 - Melt chocolate on stove
0.29 - Take off heat and let cool completely (~20 min)
0.2 - Pour into mug
0.1 - Simmer for 5 minutes
…

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Third variant: **Top-K** sampling

Method: At each step, filter to the top K options, then choose using random sampling from the probability distribution

Ingredients:

4oz Chocolate, 70% cocoa
1cup Milk

Steps:

1. Melt chocolate on stove
2. Slowly add milk
3. Heat until simmering

0.3 - Melt chocolate on stove
0.29 - Take off heat and let cool completely (~20 min)
0.2 - Pour into mug
0.1 - Simmer for 5 minutes
…

K = 3

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Third variant: **Top-K** sampling

Method: At each step, filter to the top K options, then choose using random sampling from the probability distribution

Ingredients:

4oz Chocolate, 70% cocoa
1cup Milk

Steps:

1. Melt chocolate on stove
2. Slowly add milk
3. Heat until simmering

Pour into mug

Take off heat and let cool completely (~20 min)

0.38 - Melt chocolate on stove
0.37 - Take off heat and let cool completely (~20 min)
0.25 - Pour into mug
0.1 - Simmer for 5 minutes
…

K = 3

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Fourth variant: **Top-P** sampling

Method: At each step, filter to the options that cover P% of the probability distribution, then choose using random sampling

Ingredients:

4oz Chocolate, 70% cocoa
1cup Milk

Steps:

1. Melt chocolate on stove
2. Slowly add milk
3. Heat until simmering

0.3 - Melt chocolate on stove
0.29 - Take off heat and let cool completely (~20 min)
0.2 - Pour into mug
0.1 - Simmer for 5 minutes
...

P = 80%

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Fourth variant: **Top-P** sampling

Method: At each step, filter to the options that cover P% of the probability distribution, then choose using random sampling

Ingredients:

4oz Chocolate, 70% cocoa
1cup Milk

Steps:

1. Melt chocolate on stove
2. Slowly add milk
3. Heat until simmering

Pour into mug

Simmer for 5 minutes

0.34 - Melt chocolate on stove
0.33 - Take off heat and let cool completely (~20 min)
0.22 - Pour into mug
0.11 - Simmer for 5 minutes
...

P = 80%

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Fifth variant: **Contrastive** sampling

Method: At each step, adjust scores based on similarity with recent outputs, then choose the highest scoring option

Ingredients:

4oz Chocolate, 70% cocoa
1cup Milk

Steps:

1. Melt chocolate on stove
2. Slowly add milk
3. Heat until simmering

0.3 -	* 0.1	ocolate on stove
0.29 -	* 1.0	off heat and let cool completely (~20 min)
0.2 -	* 1.0	to mug
0.1 -	* 1.0	for 5 minutes
…	…	

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Fifth variant: **Contrastive** sampling

Method: At each step, adjust scores based on similarity with recent outputs, then choose the highest scoring option

Ingredients:

4oz Chocolate, 70% cocoa
1cup Milk

Steps:

1. Melt chocolate on stove
2. Slowly add milk
3. Heat until simmering

0.03 - Melt chocolate on stove
0.29 - Take off heat and let cool completely (~20 min)
0.2 - Pour into mug
0.1 - Simmer for 5 minutes
…

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Fifth variant: **Contrastive** sampling

Method: At each step, adjust scores based on similarity with recent outputs, then choose the highest scoring option

Ingredients:

4oz Chocolate, 70% cocoa
1cup Milk

This is starting to mix modelling with inference

Steps:

1. Melt chocolate on stove
2. Slowly add milk
3. Heat until simmering

0.04 - Melt chocolate on stove
0.40 - Take off heat and let cool completely (~20 min)
0.27 - Pour into mug
0.14 - Simmer for 5 minutes
…

This can be combined with previous approaches

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Greedy variant comparison

Top-1	Argmax
Random	Sample - full distribution
Top-K	Sample - partial list, fixed length
Top-P	Sample - partial list, variable length
Contrastive	Adjust scores, then argmax

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Beam Search

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Core beam idea: Keep track of multiple options

Input at each step:

Fruit flies like bananas	Fruit flies like bananas	Fruit flies like bananas	Fruit flies like bananas
-	Noun	Noun Verb	Noun Verb Verb

Output so far [Greedy]:

Noun	Noun Verb	Noun Verb Verb	Noun Verb Verb Noun

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Core beam idea: Keep track of multiple options

Input at each step:

Fruit flies like bananas	Fruit flies like bananas	Fruit flies like bananas	Fruit flies like bananas
-	Noun	Noun Verb	Adj Noun Verb
	Verb	Adj Noun	Noun Verb Prep
	Adj	Noun Noun	Noun Verb Verb

Output so far [Beam]:

Noun	Noun Verb	Adj Noun Verb	Adj Noun Verb Noun
Verb	Adj Noun	Noun Verb Prep	Noun Verb Prep Noun
Adj	Noun Noun	Noun Verb Verb	Adj Noun Verb Adj

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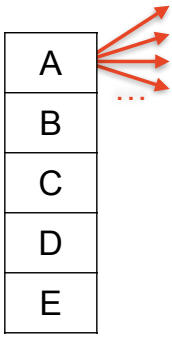
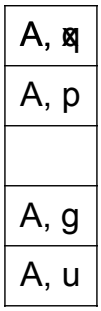
Implement with a list of options at each step that you consider extending

K best options so far (K = 5 here)



New K best options after doing this step



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Implement with a list of options at each step that you consider extending

K best options so far (K = 5 here)

A
B
C
D
E

New K best options after doing this step

A, x
A, q
B, x
A, g
B, p

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Implement with a list of options at each step that you consider extending

K best options so far (K = 5 here)

A
B
C
D
E

New K best options after doing this step

E, q
A, x
A, q
B, x

Depending on your scoring method, sometimes you can stop early - if you know that none of the remaining options will be better

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Time complexity

Number of steps

Size of beam, K

Number of options to consider (how can I extend this item?)

Beam update process (where in the list does the item go?)

Complexity = |steps| * |beam| * |options| * update

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Implementation note - how to select the top K when considering n items?

A
B
C
D
E

Simple approach, O(n * k)
For each option, go through the list to find where it goes. Once found, insert and update.

Heap approach, O(n * log k)
Use a min-heap. Update it with each new item.

Quickselect approach, O(n)
Record all options. Use quickselect to find the Kth best. Make one pass through the list to get the other K-1.

Usually k is small enough that any of these are fine and the computation of different options dominates anyway

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For variable length outputs, there are multiple possible beam definitions

Beams based on number of length of text:

Ingredients:
4oz Chocolate, 70% cocoa
1cup Milk
Chocolate shavings
Steps:
1. Melt chocolate on stove

Ingredients:
4oz Chocolate, 70% cocoa
1cup Milk
Steps:
1. Melt chocolate on stove
2. Slowly add milk

Beams based on how many recipe steps were generated:

Ingredients:
4oz Chocolate, 70% cocoa
1cup Milk
Chocolate shavings
Steps:
1. Melt chocolate on stove

Ingredients:
4oz Chocolate, 70% cocoa
1cup Milk
Steps:
1. Melt chocolate on stove

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Sometimes beam search does not capture useful variation

Ingredients:
4oz Chocolate, 70% cocoa

Ingredients:
4oz Chocolate, 75% cocoa

Ingredients:
3.5oz Chocolate, 70% cocoa

Ingredients:
3.5oz Chocolate, 75% cocoa

Ingredients:
4oz Cocoa Powder

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Graph Search

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We can also treat the problem as a search task and use graph theory

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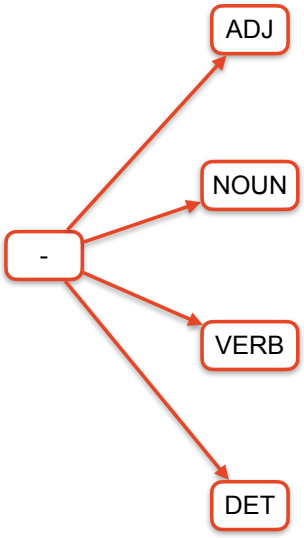
We can also treat the problem as a search task and use graph theory



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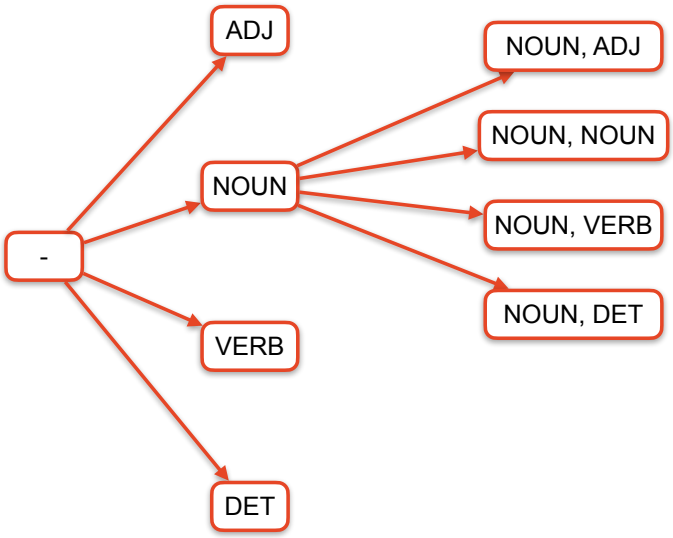

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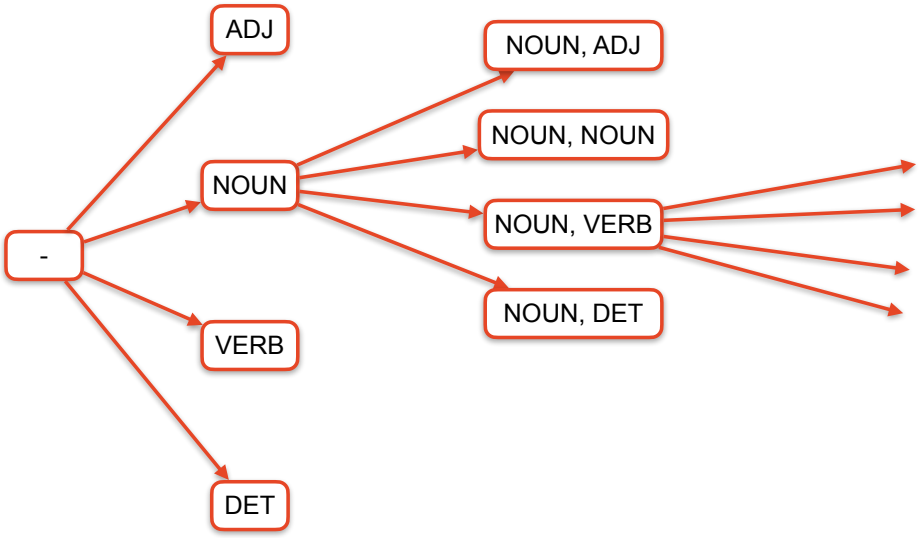
We can also treat the problem as a search task and use graph theory



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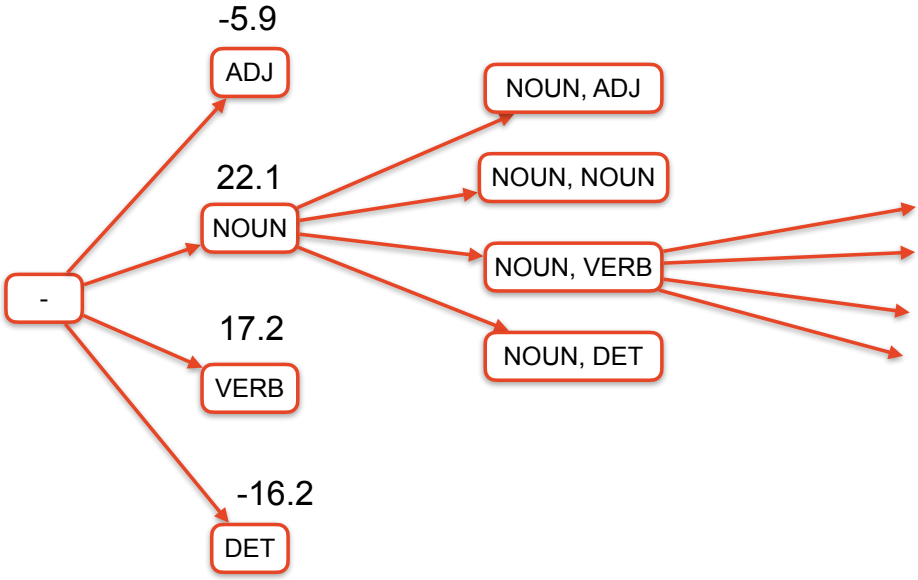
Core idea: Estimate final score and use that to guide generation



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Core idea: Estimate final score and use that to guide generation

Ingredients:
4oz Chocolate, 70% cocoa
1cup Milk
Chocolate shavings

Steps:
1. Melt chocolate on stove
2. Slowly add milk
3. Heat until simmering

Score of generation so far

Estimate of score to finish

Note - if you learned about A* search, this sounds similar, but there we are finding the shortest path, here we want the largest score.

3 minute Break - stretch and visit Menti

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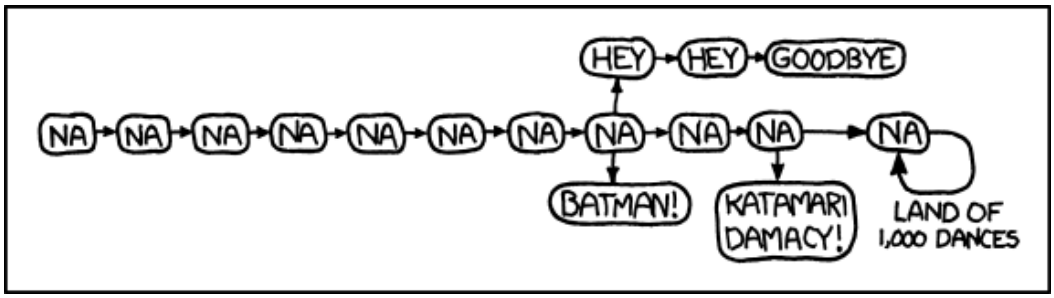
Na



[I hear that there are actual lyrics later on in Land of 1,000 Dances, but other than the occasional "I said," I've never listened long enough to hear any of them.]

Source: https://xkcd.com/851/

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Sequence Tagging

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We will consider various sequences and tags

B-PER	O	O
Joe	likes	trucks

Noun	Verb	Noun
Joe	likes	trucks

Sequences are a fixed, known, length
Tags are from a small-ish, fixed, set

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Independent predictions can be inconsistent

Possible independent predictions:

B-ORG	I-ORG	O	B-LOC	O	O	O	…

True labels:

B-ORG	I-ORG	I-ORG	I-ORG	O	O	O	…

The University of Sydney is located in …

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Greedy predictions can make mistakes that become clear too late

Predictions:

Det	Adj.	Noun	Det	Noun

True labels:

Det	Noun	Verb	Det	Noun
The	old	man	the	boat

A greedy method can't go back and undo the choices Adj & Noun!

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Considering each tag sequence separately is not feasible

17 tags in Universal Dependencies

Options = $|\text{tags}|^{|\text{words}|}$

For the last example,

17^5 = 1,419,857 options

For a 20 word sentence (average length in news):

17^{20} = 4,064,231,406,647,572,522,401,601
(4.1 Septillion)

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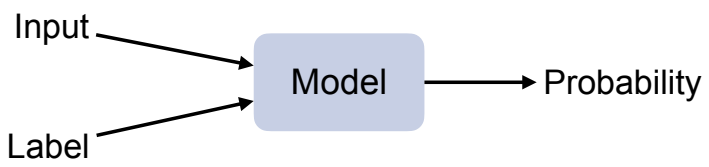
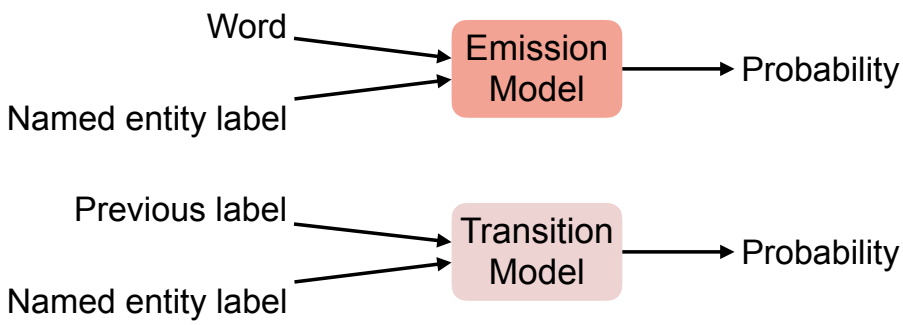
For certain models we can find the highest scoring sequence efficiently



Our model needs to calculate the score in pieces of two types:



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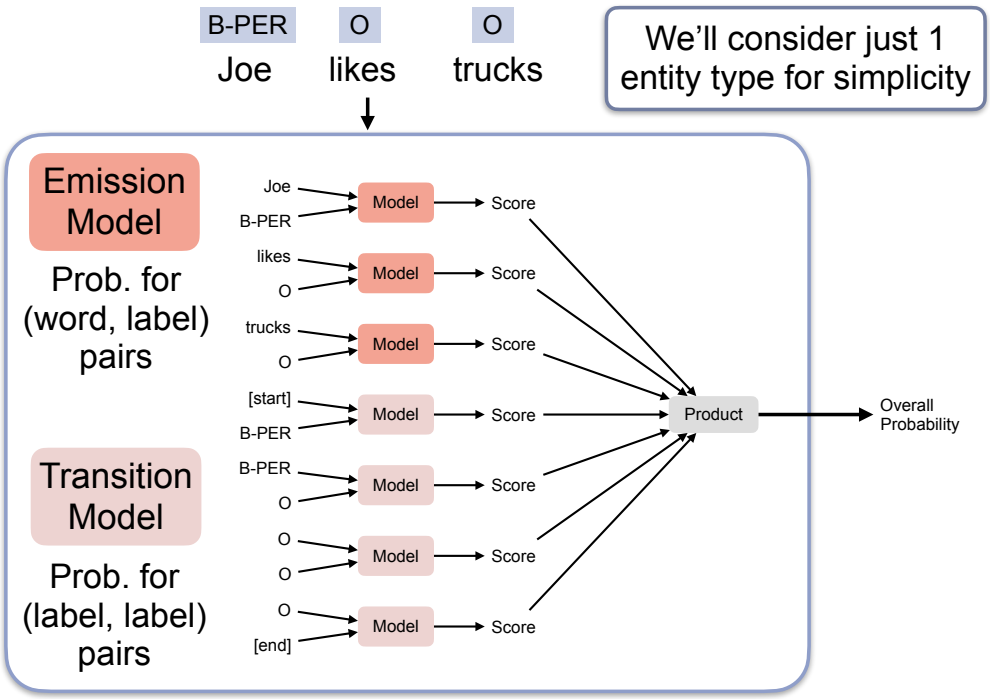
We can still think of it as a model for whole sequences



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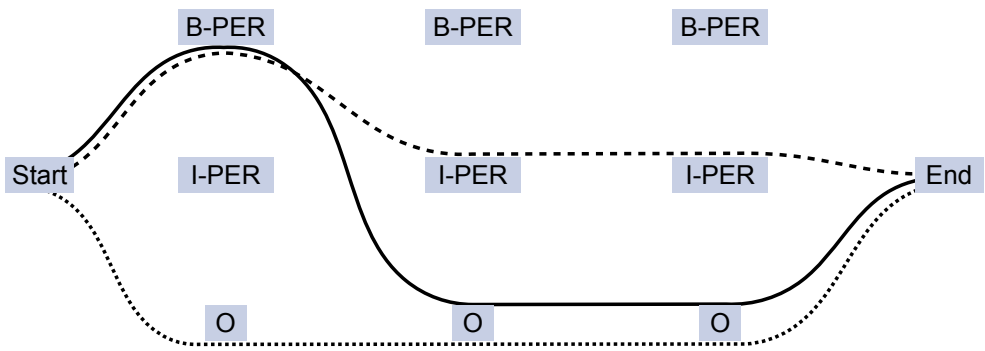
To understand the algorithm, we can draw paths through a grid of labels



Joe	likes	trucks	——
I	like	chocolate	·····
Jonathan	Kay	Kummerfeld	- - - -

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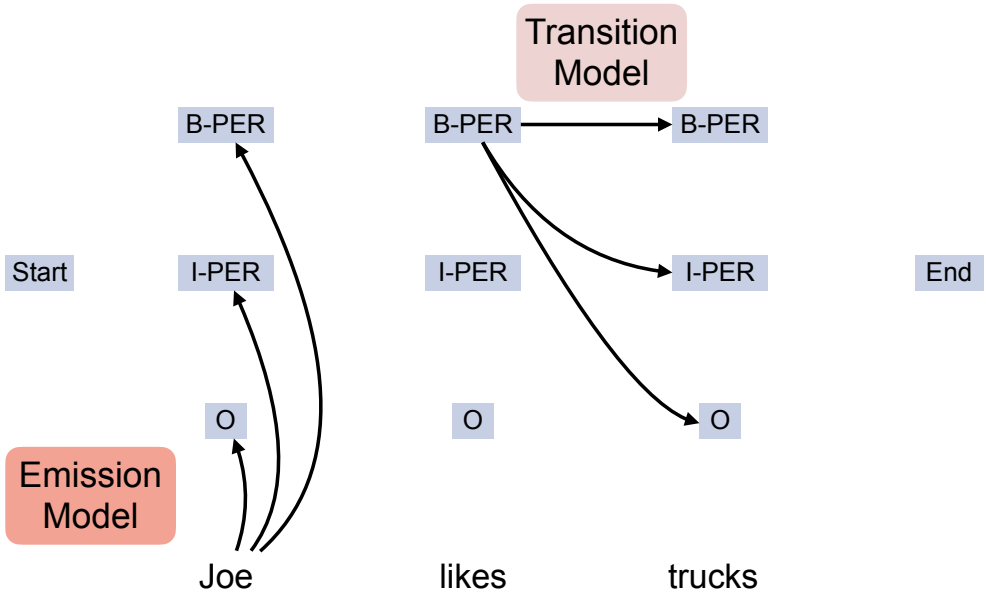
The two model parts from earlier correspond to parts of this grid



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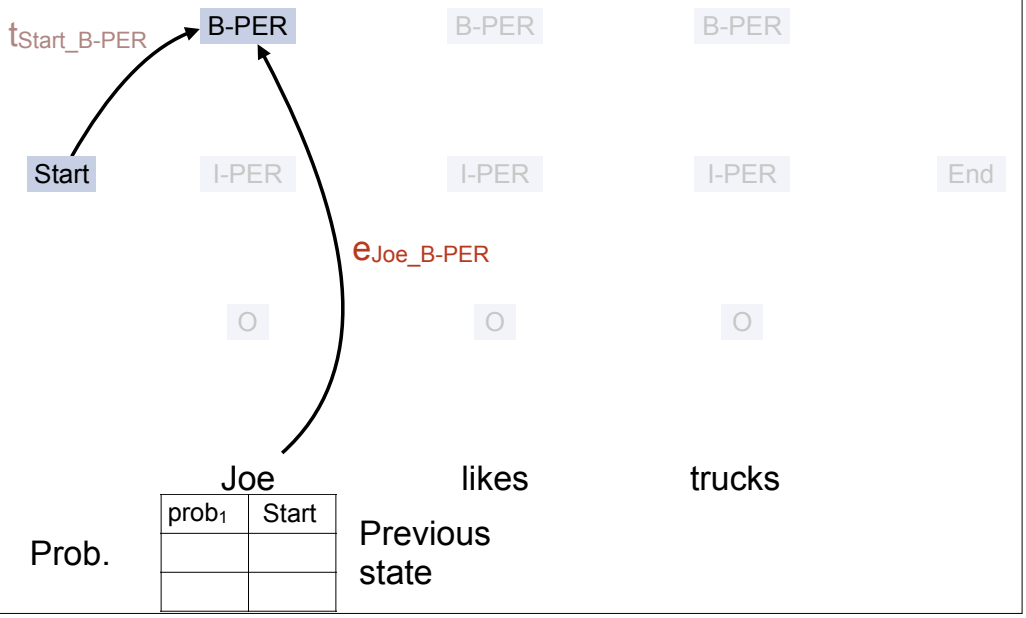
We calculate intermediate scores, first for each possible start

$$prob_1 = e_{Joe_B\text{-}PER} * t_{Start_B\text{-}PER}$$



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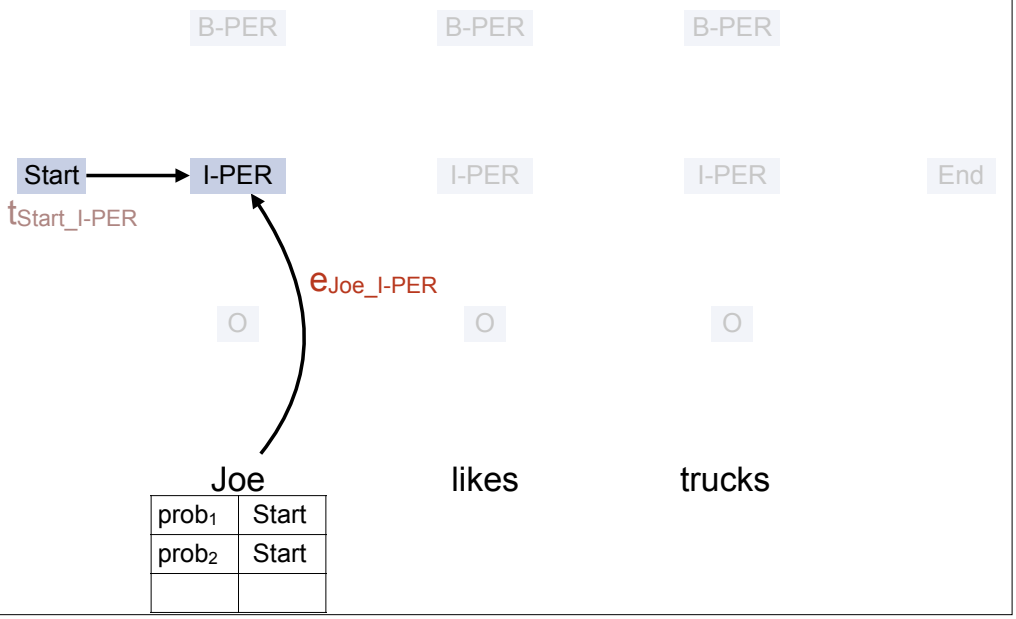
We calculate intermediate scores, first for each possible start

$$prob_2 = e_{Joe_I\text{-}PER} * t_{Start_I\text{-}PER}$$



$prob_1$	Start
$prob_2$	Start

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We calculate intermediate scores, first for each possible start

$$prob_3 = e_{Joe_O} * t_{Start_O}$$

$prob_1$	Start
$prob_2$	Start
$prob_3$	Start

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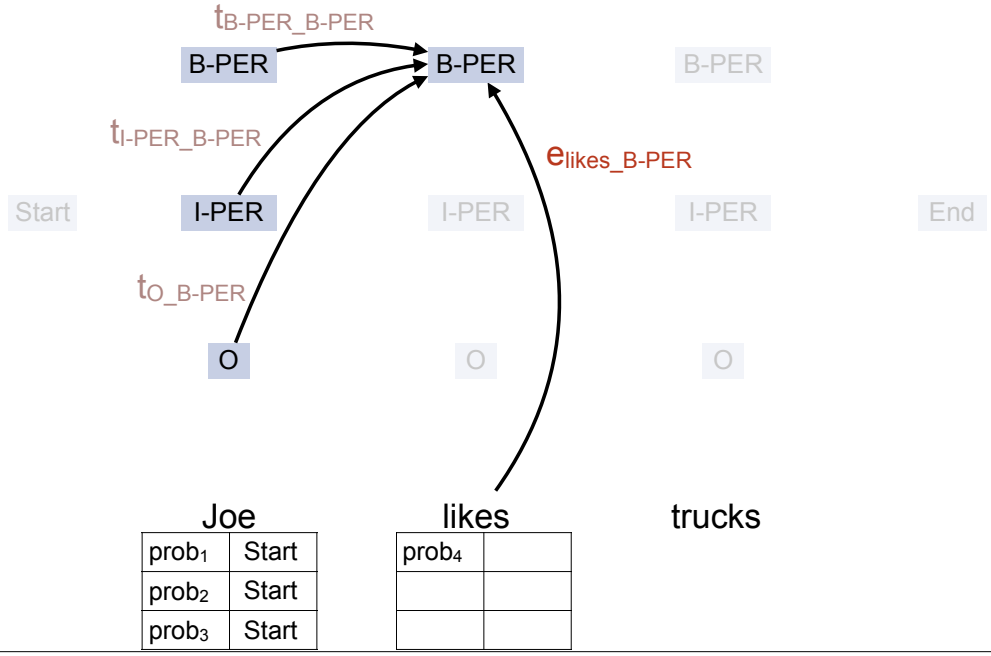
We calculate intermediate scores, then continuations, noting the best

$$prob_4 = e_{likes_B\text{-}PER} * \max(t_{prev_B\text{-}PER} * prob_{prev})$$



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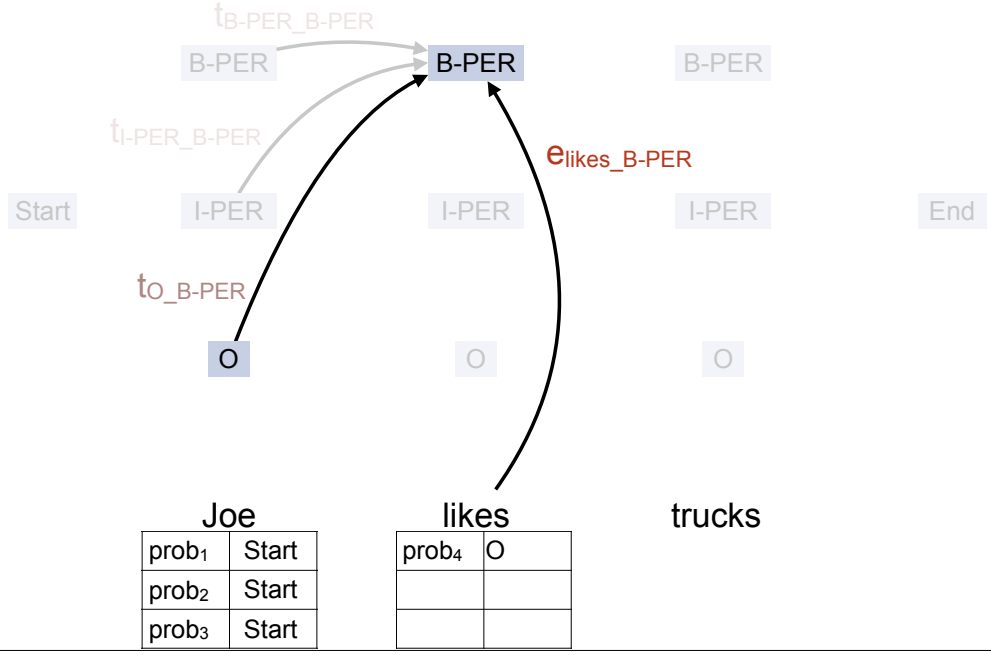

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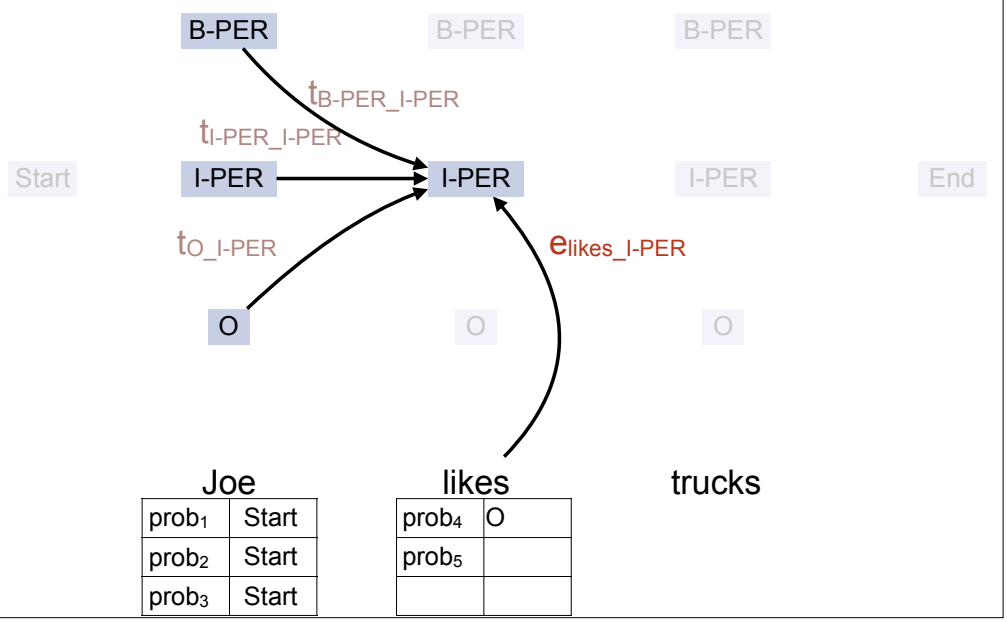
We calculate intermediate scores, then continuations, noting the best

$$prob_5 = e_{likes_I\text{-}PER} * \max(t_{prev_I\text{-}PER} * prob_{prev})$$



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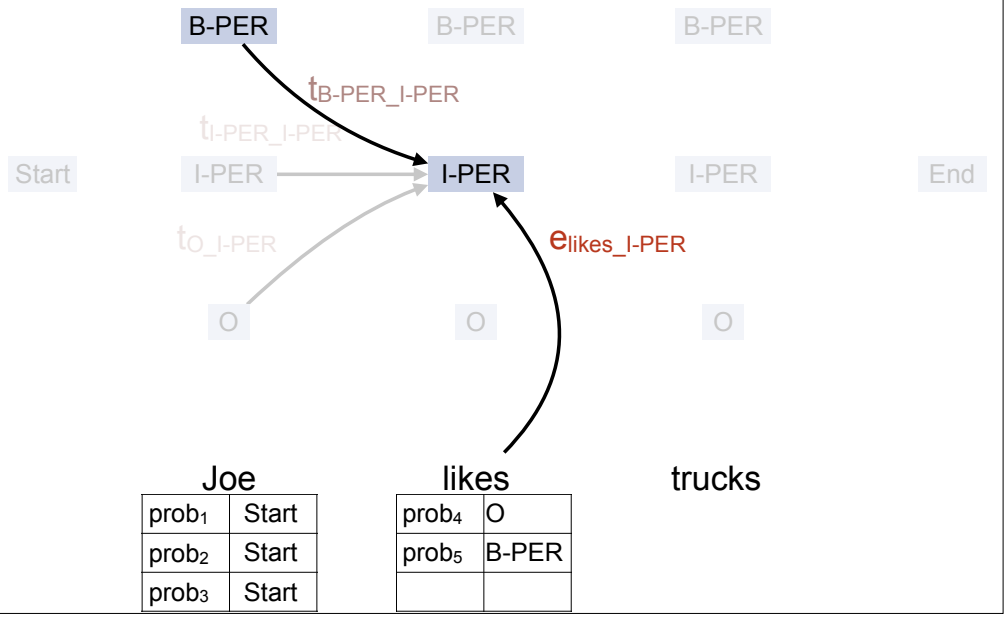
$$prob_5 = e_{likes_I\text{-}PER} * \max(t_{prev_I\text{-}PER} * prob_{prev})$$



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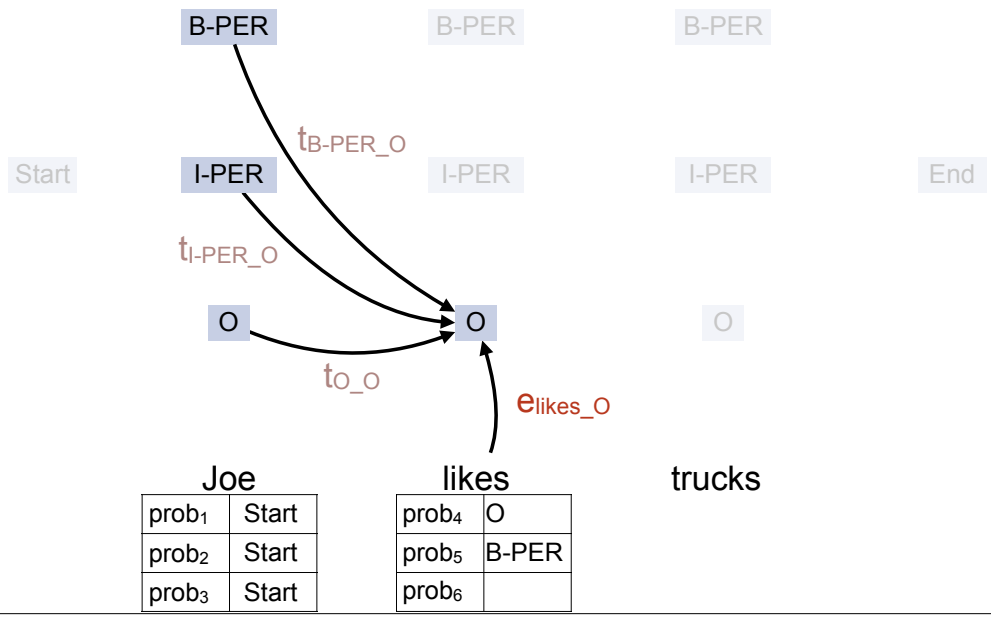
We calculate intermediate scores, then continuations, noting the best

$$prob_6 = e_{likes_O} * \max(t_{prev_O} * prob_{prev})$$



$prob_1$	Start
$prob_2$	Start
$prob_3$	Start

$prob_4$	O
$prob_5$	B-PER
$prob_6$	

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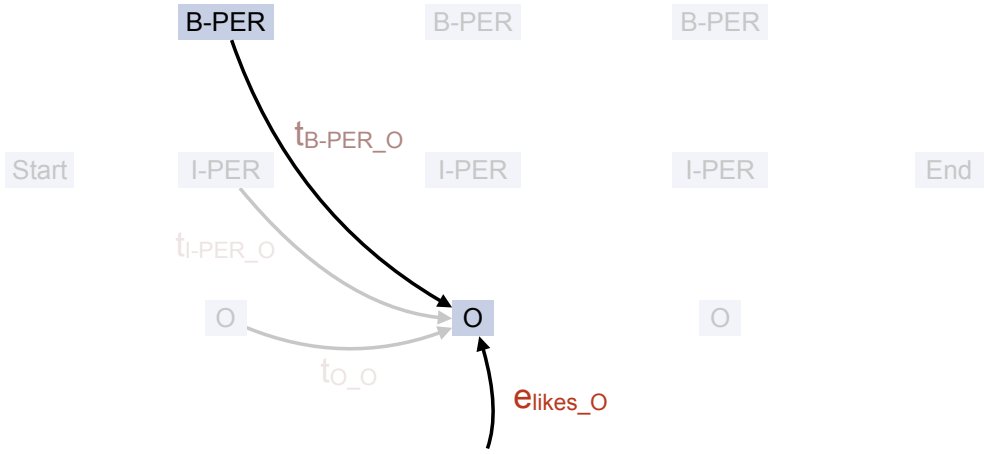
$$prob_6 = e_{likes_O} * \max(t_{prev_O} * prob_{prev})$$



$prob_1$	Start
$prob_2$	Start
$prob_3$	Start

$prob_4$	O
$prob_5$	B-PER
$prob_6$	B-PER

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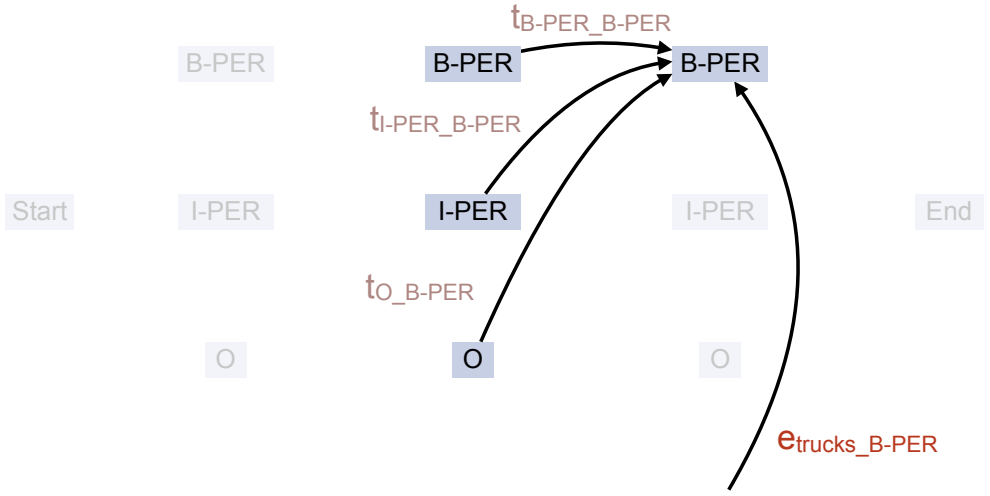
Each step only looks back one step

$$prob_7 = e_{trucks_B\text{-}PER} * \max(t_{prev_B\text{-}PER} * prob_{prev})$$



Joe	
$prob_1$	Start
$prob_2$	Start
$prob_3$	Start

likes	
$prob_4$	O
$prob_5$	B-PER
$prob_6$	B-PER

trucks	
$prob_7$	

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Each step only looks back one step

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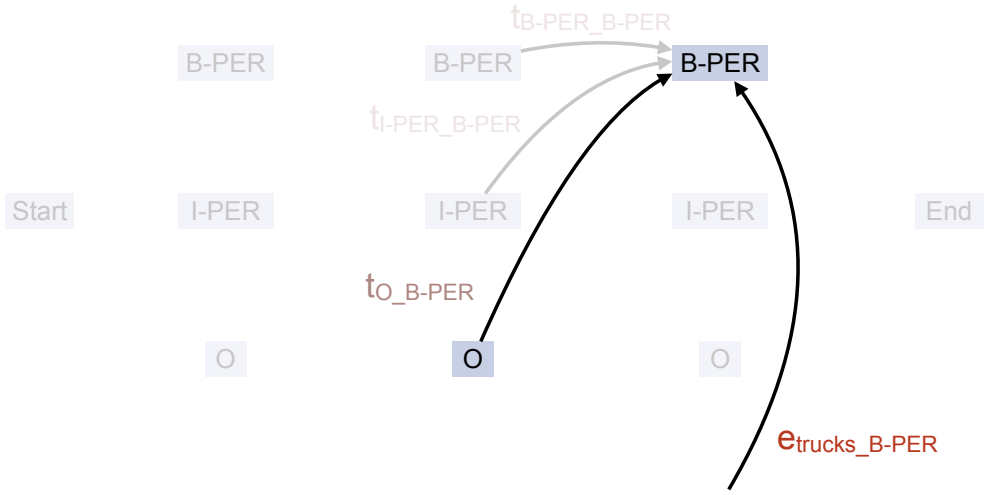

Joe	
$prob_1$	Start
$prob_2$	Start
$prob_3$	Start

likes	
$prob_4$	O
$prob_5$	B-PER
$prob_6$	B-PER

trucks	
$prob_7$	O

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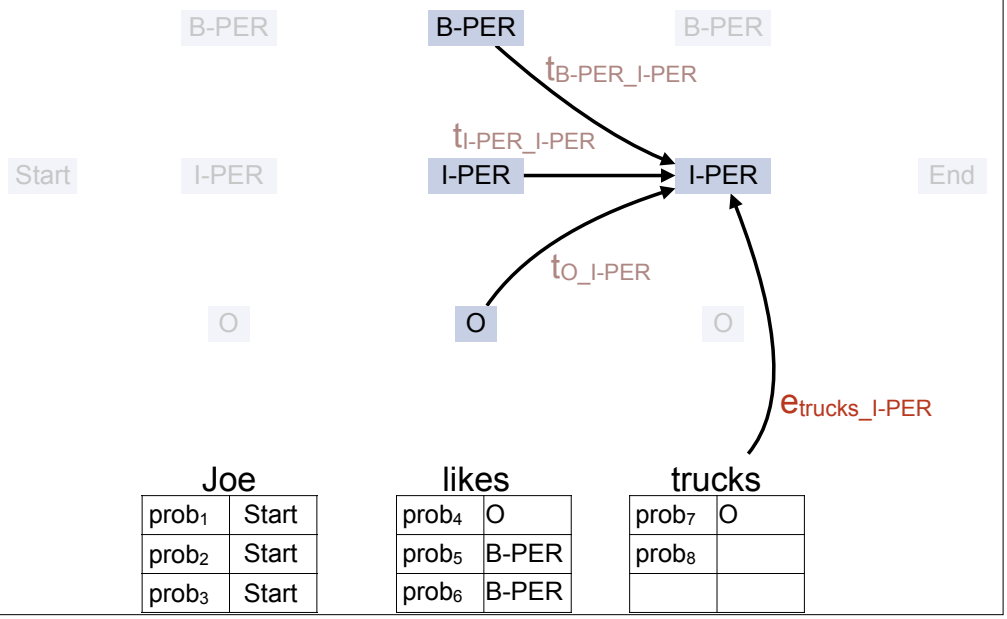
$$prob_8 = e_{trucks_I\text{-}PER} * \max(t_{prev_I\text{-}PER} * prob_{prev})$$



$prob_1$	Start
$prob_2$	Start
$prob_3$	Start

$prob_4$	O
$prob_5$	B-PER
$prob_6$	B-PER

$prob_7$	O
$prob_8$	

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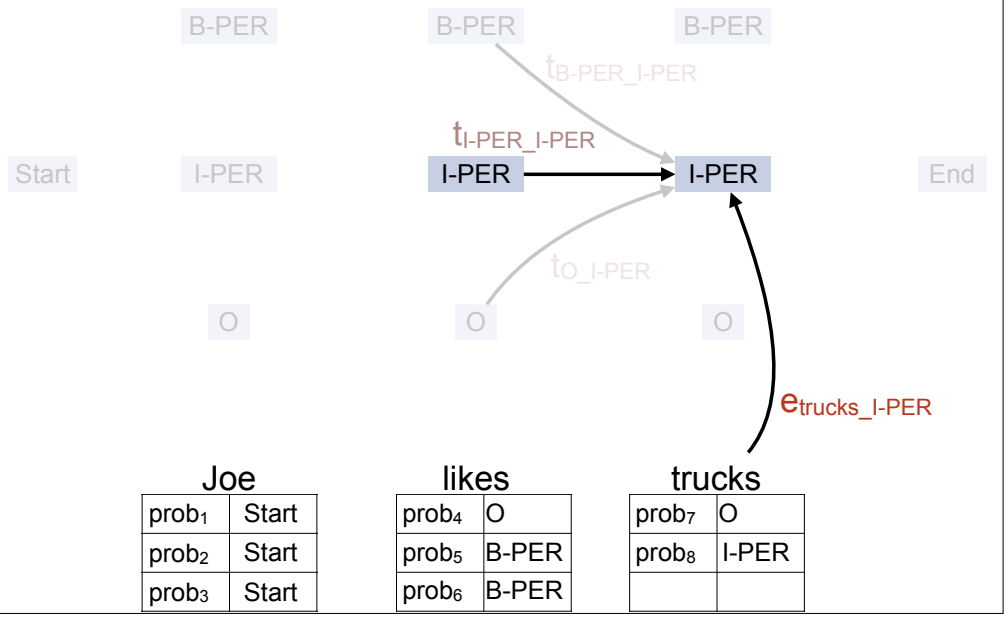
$$prob_8 = e_{trucks_I\text{-}PER} * \max(t_{prev_I\text{-}PER} * prob_{prev})$$



$prob_1$	Start
$prob_2$	Start
$prob_3$	Start

$prob_4$	O
$prob_5$	B-PER
$prob_6$	B-PER

$prob_7$	O
$prob_8$	I-PER

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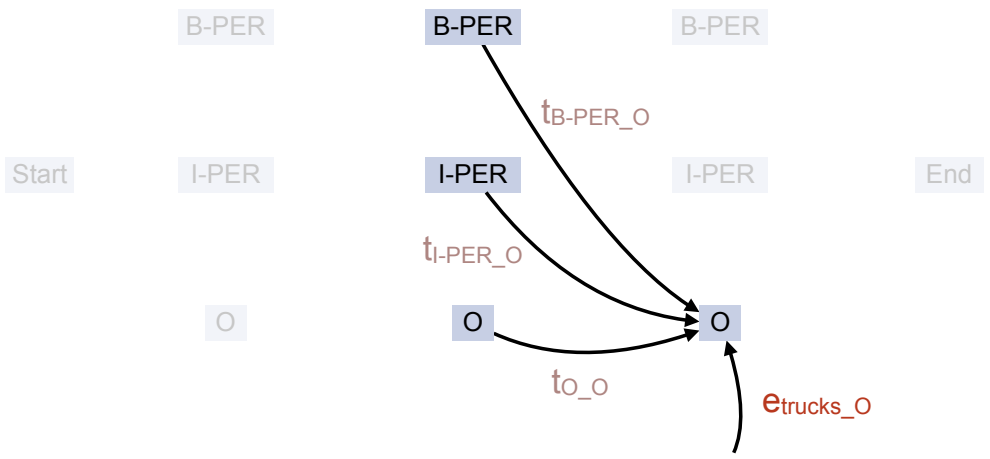
$$prob_9 = e_{trucks_I\text{-}PER} * \max(t_{prev_I\text{-}PER} * prob_{prev})$$



Joe	
$prob_1$	Start
$prob_2$	Start
$prob_3$	Start

likes	
$prob_4$	O
$prob_5$	B-PER
$prob_6$	B-PER

trucks	
$prob_7$	O
$prob_8$	I-PER
$prob_9$	

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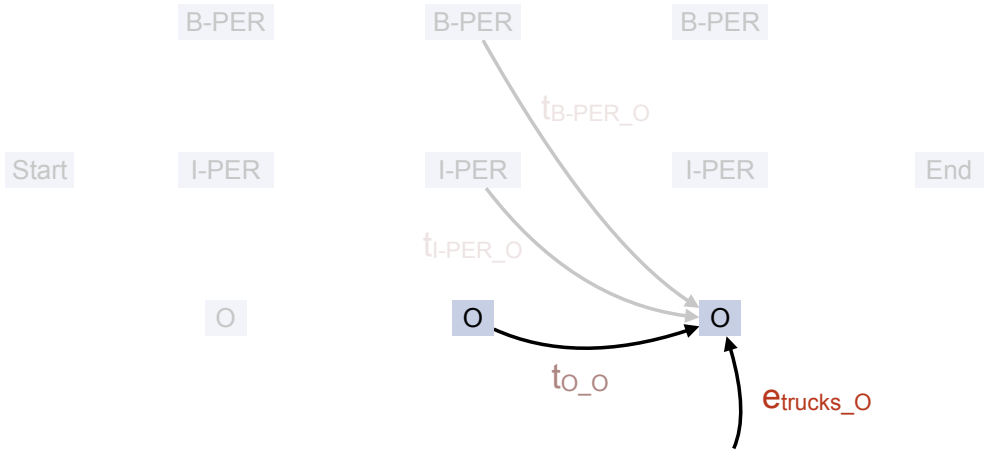
$$prob_9 = e_{trucks_I\text{-}PER} * \max(t_{prev_I\text{-}PER} * prob_{prev})$$



Joe	
$prob_1$	Start
$prob_2$	Start
$prob_3$	Start

likes	
$prob_4$	O
$prob_5$	B-PER
$prob_6$	B-PER

trucks	
$prob_7$	O
$prob_8$	I-PER
$prob_9$	O

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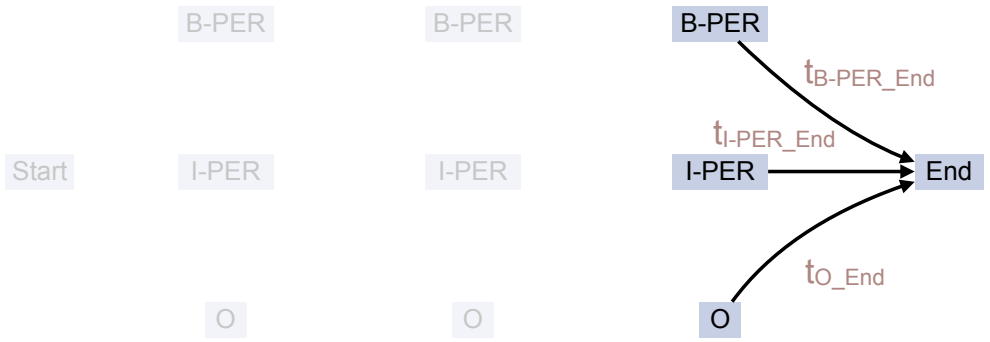
Each step only looks back one step

$$prob_{10} = \max(t_{prev_End} * prob_{prev})$$



Joe	
$prob_1$	Start
$prob_2$	Start
$prob_3$	Start

likes	
$prob_4$	O
$prob_5$	B-PER
$prob_6$	B-PER

trucks	
$prob_7$	O
$prob_8$	I-PER
$prob_9$	O

$prob_{10}$	

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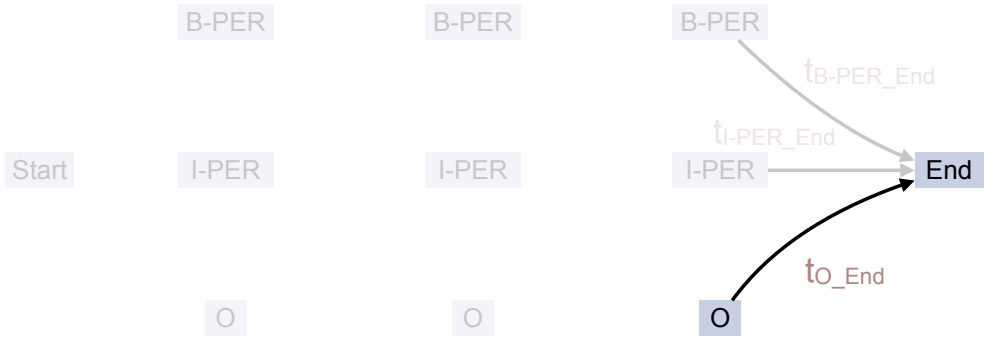
$$prob_{10} = \max(t_{prev_End} * prob_{prev})$$



Joe	
$prob_1$	Start
$prob_2$	Start
$prob_3$	Start

likes	
$prob_4$	O
$prob_5$	B-PER
$prob_6$	B-PER

trucks	
$prob_7$	O
$prob_8$	I-PER
$prob_9$	O

$prob_{10}$	O

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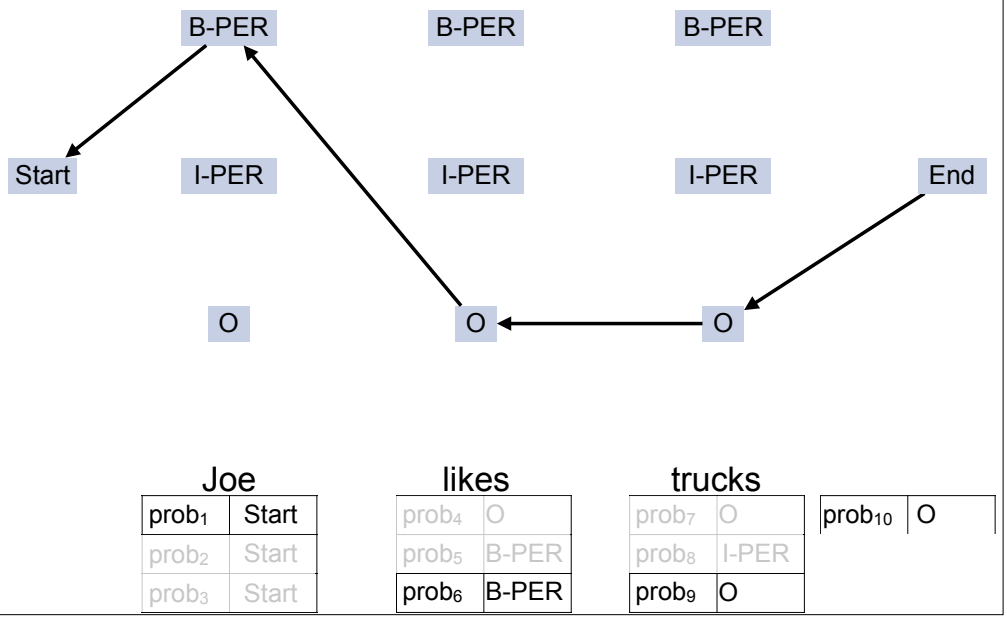
To get the path, follow the references backwards to the start



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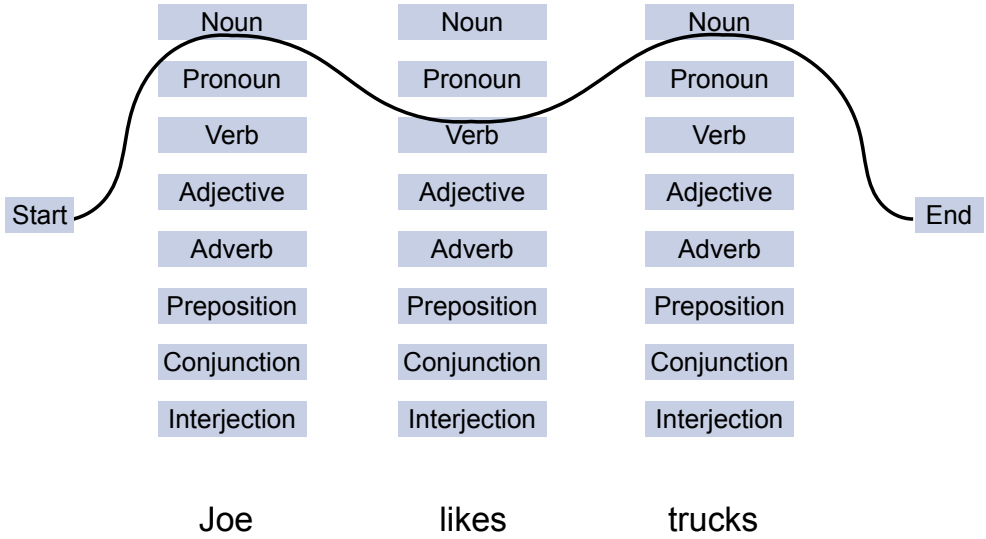
We can apply the same idea to POS tagging



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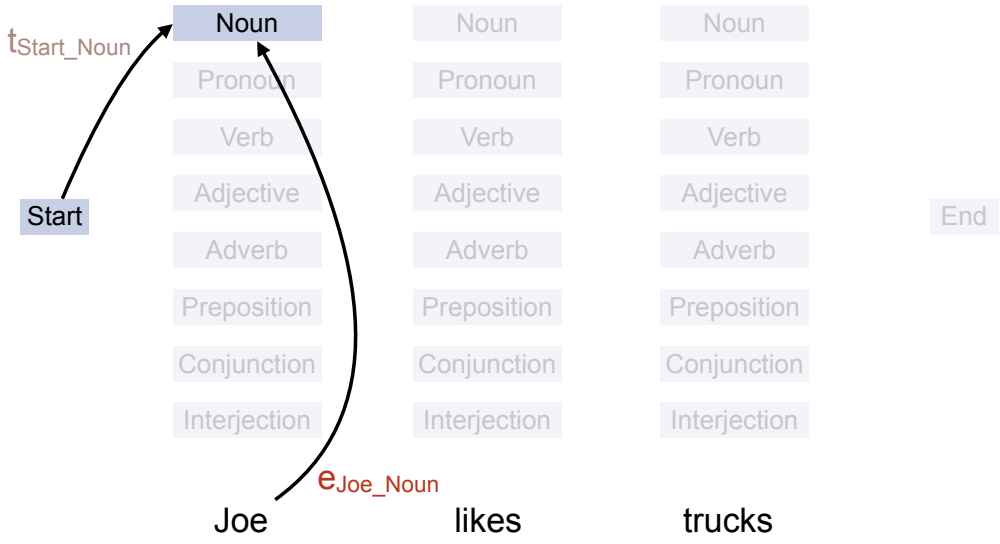

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Pseudocode for the algorithm to calculate scores and previous labels

```
Create storage for best option for each
(position, label) pair

For each position in input:
    For each label:
        Best score = -infinity
        Best previous = None
        For each previous_label:
            Score = previous_label's prob *
                    prob(label | input and
                    previous label)
            If score is better than best:
                Update best
        Save best score and best previous for
        this (position, label) pair
```

Called the Viterbi algorithm

(note, also need to account for start and end)

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Pseudocode for the algorithm to get the path

```
Path = []
Current label = End
For position from end of sequence to start:
    Current label = Best for this position and
                    the current label
    Append current label to Path

Reverse path
```


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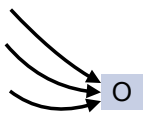
Time complexity is $O(|words| * |labels|^2)$

Joe, likes, trucks

|words| steps

B-PER
I-PER
O
likes

In each step, consider |labels|



For each option, consider
|labels| previous options

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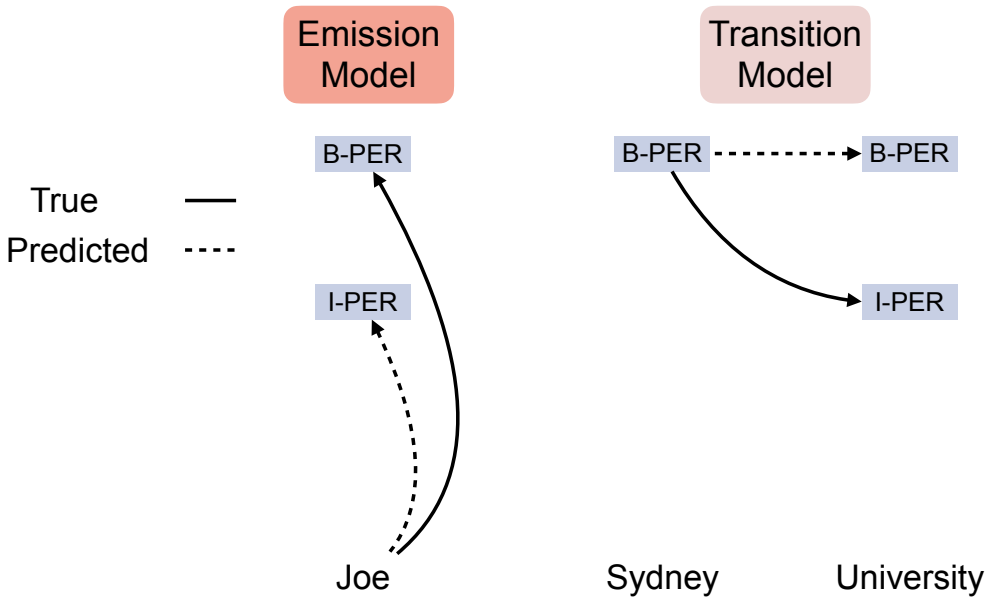
Learn by comparing the final output with the true answer



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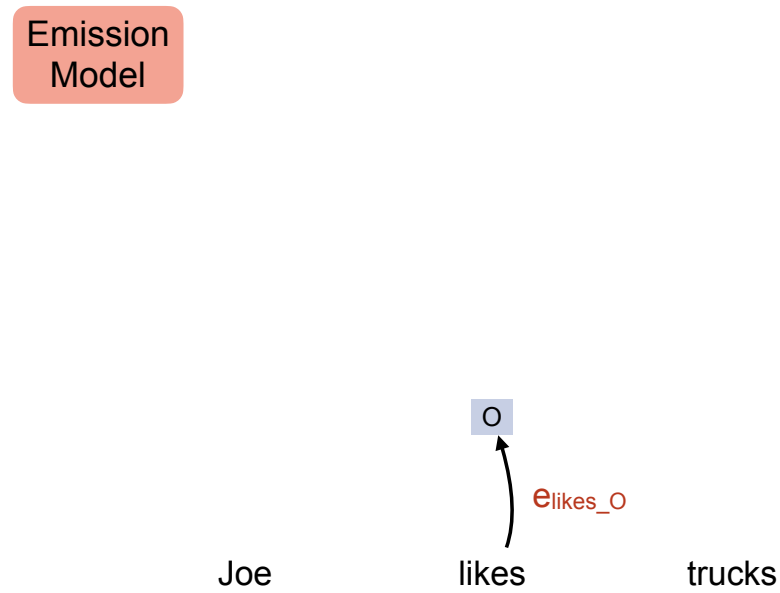
The model can use all of the input, and the previous label



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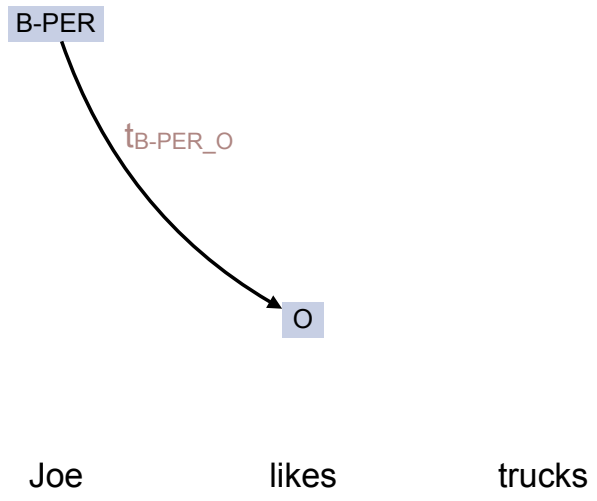
The model can use all of the input, and the previous label

Transition Model



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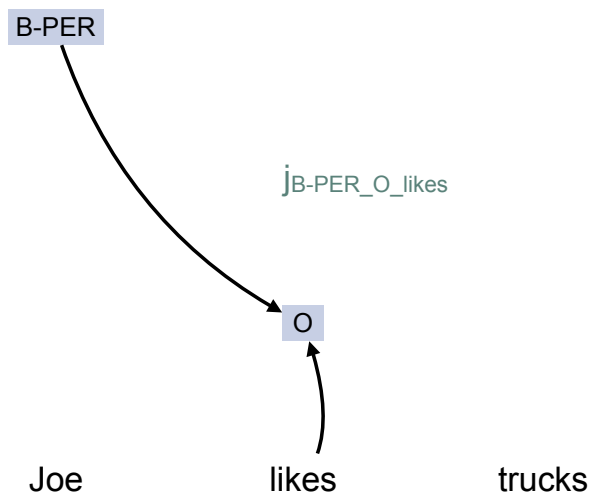
We can also define a model that does both scores at once

Joint Model

$score_{current} = max(j_{prev_label_word} * score_{prev})$



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If we are willing to increase complexity, we can use more label context

Previous_Current

		B-PER_B-PER	B-PER_B-PER	
		I-PER_B-PER	I-PER_B-PER	
	Start_B-PER	O_B-PER	O_B-PER	
		B-PER_I-PER	B-PER_I-PER	
Start	Start_I-PER	I-PER_I-PER	I-PER_I-PER	End
		O_I-PER	O_I-PER	
	Start_O	B-PER_O	B-PER_O	
		I-PER_O	I-PER_O	
		O_O	O_O	
	Joe	likes	trucks	

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If we are willing to increase complexity, we can use more label context

B-PER_B-PER	B-PER_B-PER
I-PER_B-PER	I-PER_B-PER
O_B-PER	O_B-PER
B-PER_I-PER	B-PER_I-PER
I-PER_I-PER	I-PER_I-PER
O_I-PER	O_I-PER
B-PER_O	B-PER_O
I-PER_O	I-PER_O
O_O	O_O
likes	trucks

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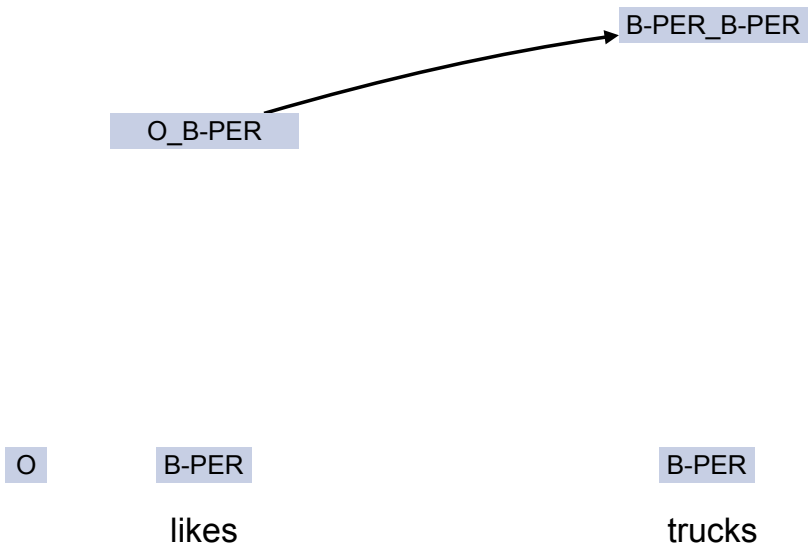

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Now, time complexity is $O(|words| * |labels|^3)$

Joe, likes, trucks

|words| steps

In each step, consider |labels|*|labels|

For each option, consider |labels| previous options

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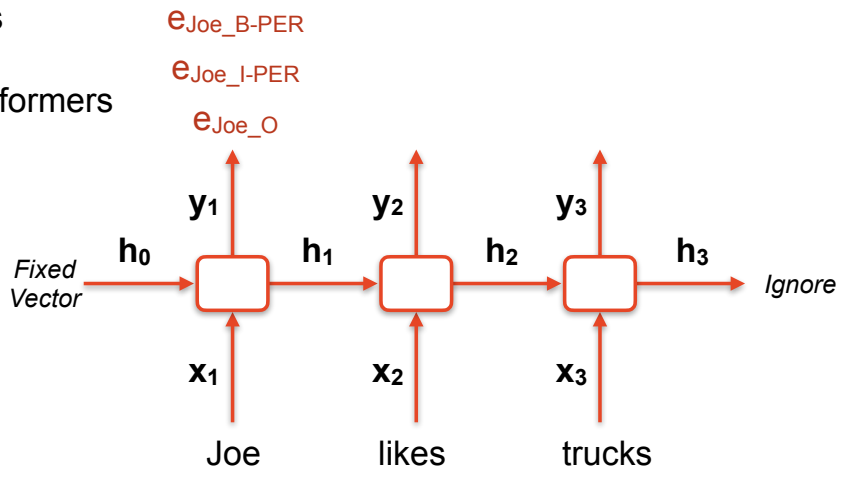
We can apply this method to a variety of models

Linear models (e.g., CRF)

Feedforward Networks

RNNs

Transformers



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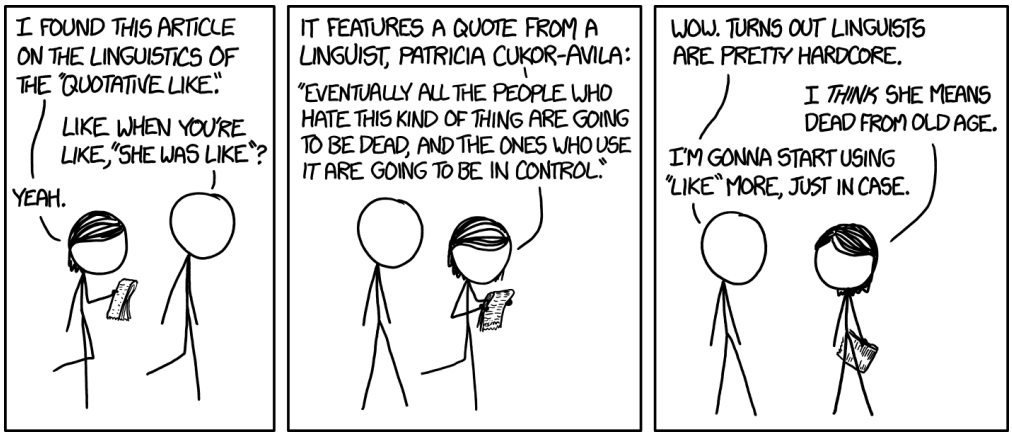
Quotative Like



[God was like, "Let there be light," and there was light.]

Source: https://xkcd.com/1483/

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Graph Parsing

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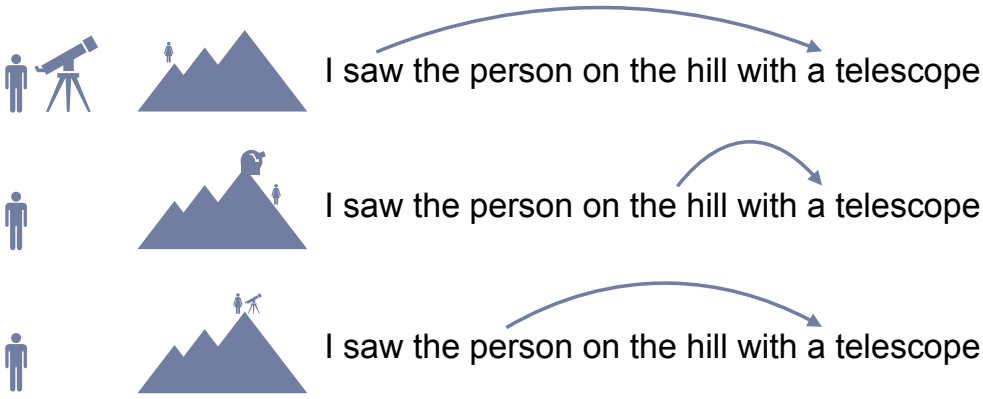
Syntactic parsing can resolve ambiguities

I saw the person on the hill with a telescope



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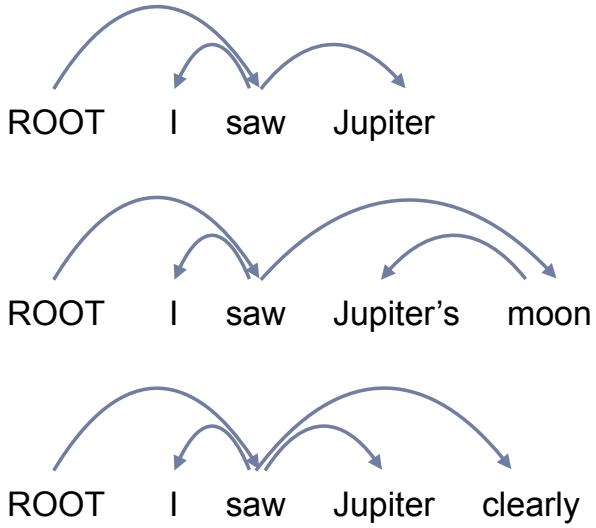
Arcs / Edges / Dependencies go from a head to a dependent



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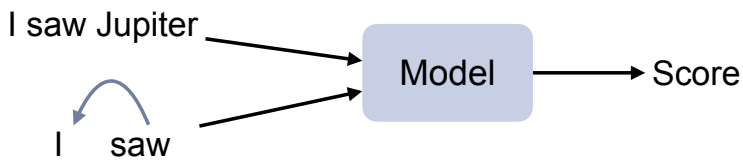
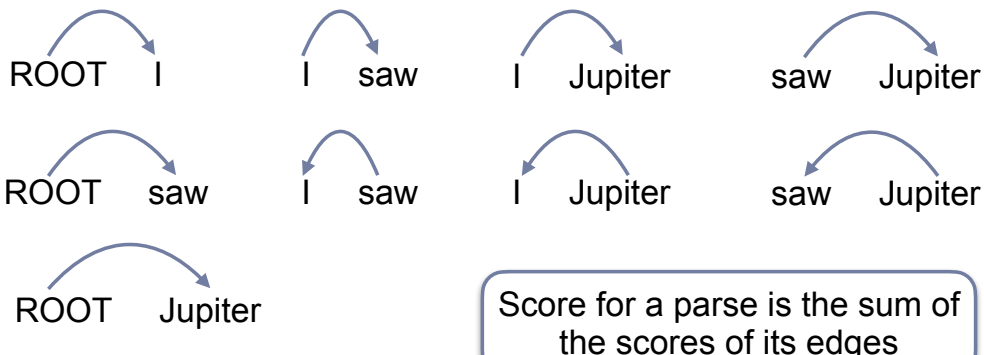
We'll use an arc model to score options





Score for a parse is the sum of the scores of its edges

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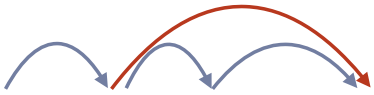
Requirements for our inference method

1. Produce a tree



2. No crossing arcs

3. A single root

4. The tree has the highest score of all possible trees

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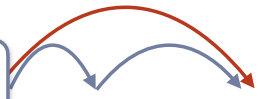
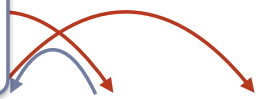


In my research I invented an algorithm that overcomes these two constraints

2. No crossing arcs



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References are when entities are mentioned in text

Victoria Chen, CFO of Megabucks Banking, saw her pay jump to $2.3 million, as the 38-year-old became the company's president. It is widely known that she came to Megabucks from rival Lotsabucks.

Example text is from the J+M textbook, chapter 23

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Coreference is identifying which mentions refer to the same entity

Victoria Chen

Megabucks Banking

Lotsabucks

her pay

Victoria Chen, CFO of Megabucks Banking, saw her pay jump to $2.3 million, as the 38-year-old became the company's president. It is widely known that she came to Megabucks from rival Lotsabucks.

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Entity Linking / Wikification is a related task that links each entity to an external knowledge base

Victoria Chen

Megabucks Banking

Lotsabucks

her pay

Victoria Chen, CFO of Megabucks Banking, saw her pay jump to $2.3 million, as the 38-year-old became the company's president. It is widely known that she came to Megabucks from rival Lotsabucks.

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The search space is huge!

Possible references:
|words| * |words|

Possible sets:
All ways to cluster
all references

Victoria Chen, CFO of Megabucks Banking, saw her pay jump to $2.3 million, as the 38-year-old became the company's president. It is widely known that she came to Megabucks from rival Lotsabucks.

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First idea for simplification: do mention detection and filtering first

Possible references:
|words| * |words|

Victoria
Victoria Chen
Victoria Chen ,
Victoria Chen , CFO
Victoria Chen , CFO of
Victoria Chen , CFO of Megabucks
Victoria Chen , CFO of Megabucks Banking
Victoria Chen , CFO of Megabucks Banking ,
Victoria Chen , CFO of Megabucks Banking , saw
Victoria Chen , CFO of Megabucks Banking , saw her

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Possible references:
|words| * |words|

Victoria
Victoria Chen
~~Victoria Chen ,~~
Victoria Chen , CFO
~~Victoria Chen , CFO of~~
~~Victoria Chen , CFO of Megabucks~~
~~Victoria Chen , CFO of Megabucks Banking~~
~~Victoria Chen , CFO of Megabucks Banking ,~~
~~Victoria Chen , CFO of Megabucks Banking , saw~~
~~Victoria Chen , CFO of Megabucks Banking , saw her~~

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Second idea for simplification: Link pairs of mentions

Just find the best previous mention for this mention to be coreferent with

Victoria Chen, CFO of Megabucks Banking, saw her pay jump to $2.3 million, as the 38-year-old became the company's president. It is widely known that she came to Megabucks from rival Lotsabucks.

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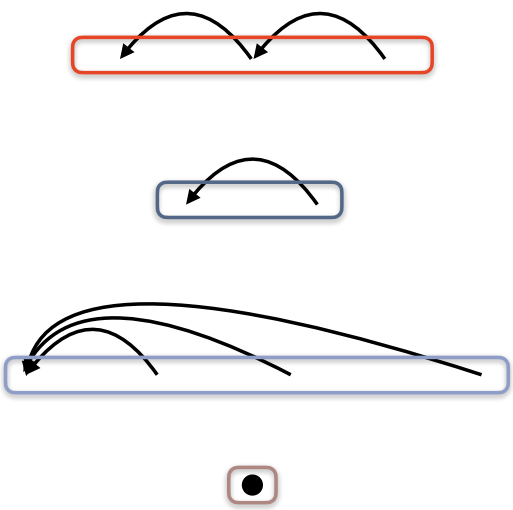
Third idea for simplification: Find the transitive closure



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But what about fancy inference algorithms?

Sometimes the simple idea ***does*** work!

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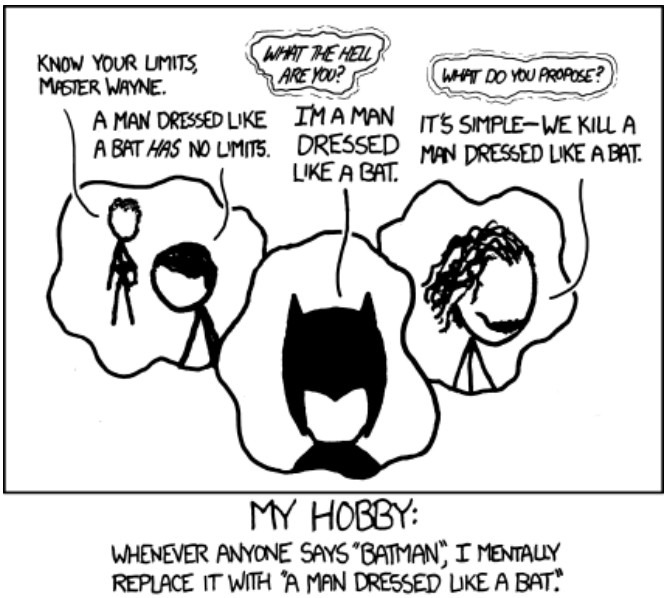

Batman

[I'm really worried Christopher Nolan will kill a man dressed like a bat in his next movie. (The man will be dressed like a bat, I mean. Christopher Nolan won't be, probably.)]

Source: https://xkcd.com/1004/

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Implement an RNN in PyTorch

Pre-work:	Reading and example RNN code
Workshop:	Implementing the RNN from lecture 3

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Muddy Card

Open now, closes Thursday 11:59pm

Step 1: Enroll in Canvas (if you haven't already)
https://canvas.sydney.edu.au/enroll/LWPKRW

Step 2: Complete week 3 muddy card
https://canvas.sydney.edu.au/courses/72300/quizzes/332400

If you do not wish to participate in the study,
use the Microsoft form instead
Go to Ed → Lessons → Muddy Cards → Form